SafetyNet PV — Pharmacovigilance Platform Assessment

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1. Executive summary

SafetyNet is Vernova's **pharmacovigilance (PV) case management platform**, used by Global Patient Safety to intake, triage, assess, and submit **Individual Case Safety Reports (ICSRs)** for all marketed products across every operating region. The platform ingests adverse event reports from HCPs, patients, literature screening, and partner licensees; routes them for medical review; and submits regulatory **E2B(R3)** XML to FDA FAERS, EMA EudraVigilance, PMDA, Health Canada, TGA, and 14 other agencies.

The system is a vendor Oracle Argus Safety 7.0 deployment (commissioned 2014) with extensive in-house customisations: a Siebel-based intake portal, custom E2B(R3) gateway code, and a bespoke literature-screening workflow. The Argus application tier runs on Windows Server 2012 R2; the database is Oracle 11g R2. The gateway and portal run on JBoss EAP 6.4 on RHEL 6.

PV is Vernova's most regulatorily-exposed system. Missed or late submissions are directly inspectable; the EU-QPPV is personally liable under GVP Module I. An outage during a submission deadline is a reportable event.

SafetyNet is on the mandatory FY27 remediation list driven by three forces: Oracle 11g R2 and WebLogic/JBoss being out of support, an EMA Pharmacovigilance Inspection in Q2 FY27 flagging the IRT-equivalent gateway monitoring gaps, and the upcoming **ICH E2B(R3) regional profile updates** scheduled for H2 FY27 that will require application changes anyway.

Headline numbers - ICSRs processed per year: ~48,000 (growing 12% YoY as marketed portfolio expands) - Active users: ~220 safety physicians, scientists, and operations staff across 14 regions - Databases: Argus Safety primary (~2.4 TB Oracle), Reports (~900 GB), Audit (~1.1 TB) - Downstream submission endpoints: 20 regulatory agencies - Release cadence: 2 to 3 months (quarterly) — compressed from 6 months in 2024 - Annual run cost: approximately €2.1M (Oracle Argus licences + Oracle DB + hosting + 24x7 managed service)

2. Business criticality and regulatory scope

Criticality: Tier-0 regulatory. An outage that prevents on-time submission of expedited (Day 15 / Day 7) ICSRs is a notifiable incident to EMA under GVP Module I §I.B.11, and comparable rules apply at FDA, PMDA, MHRA, Health Canada, TGA, and all other agencies.

- RTO: 2 hours (business hours), 4 hours (24x7).
- RPO: 15 minutes — enforced by Oracle Data Guard synchronous replication.
- Mandatory 24x7 on-call for Patient Safety operations.
- Backup EU-QPPV coverage required at all times.

Regulatory scope: - **FDA 21 CFR Part 11** — audit trail, e-signatures on case lock - **EU GMP Annex 11** — computerised systems - **EU GVP (Good Pharmacovigilance Practices) Modules I, VI, IX, XVI** — case management, signal detection, risk management plans, additional monitoring - **ICH E2B(R3)** — the regulatory messaging standard; current profile v3.0.1, regional profiles expected to update H2 FY27 - **ICH E2D** — post-approval expedited reporting - **GDPR / HIPAA** — patient-identifiable data in ICSRs - **GAMP 5** — Argus Safety classified as Cat 4 (configured product); customisations are Cat 5 - **US DSCSA / EU FMD** — not in scope for PV, mentioned here only because our serialization team uses adjacent infrastructure

3. Functional scope

- Case intake (web portal, email ingestion, direct literature pull, partner feeds)
- Automatic case duplicate detection
- Triage and assignment to case owner
- Medical assessment (seriousness, causality, expectedness)
- MedDRA auto-coding and reviewer override
- Narrative generation with templated AI-assisted drafts (added 2023; currently in review)
- Case lock with electronic signature
- E2B(R3) XML generation per regional profile
- Regulatory gateway submission + acknowledgement tracking (ACK1/ACK2/ACK3)
- Signal detection (quarterly PBRER / PSUR support, routine signal detection runs)
- Literature screening workflow (Embase + PubMed daily)
- Reconciliation with partner licensees (DSUR, PSUR exchange)
- Reconciliation with Clinical safety database (bi-directional with the Clinical Operations team's safety system)

Out of scope for SafetyNet: clinical trial SAE management (handled upstream in the Clinical Safety system, feeds in via nightly reconciliation).

4. Logical architecture

Classic Oracle Argus 7.0 three-tier with heavy customisation:

1. **Presentation tier (3 systems):** - Argus Safety thick-client (ActiveX, Windows-only, Internet Explorer 11 or Edge in IE Mode) - Siebel-based intake portal (Siebel 8.1) for external HCP/patient reporters - Custom web dashboards (ASP.NET MVC 5) for regional PV ops managers
2. **Business tier:** - Argus application services (Windows Server 2012 R2) - E2B gateway on JBoss EAP 6.4 (RHEL 6), handling bi-directional submission + ACK - Literature screening background workers (Python 3.6 + Celery, on separate Linux VMs)
3. **Data tier:** - Oracle 11g R2 RAC (primary + Data Guard DR) - MedDRA dictionary database (same Oracle instance, separate schema; refreshed semi-annually)

5. Physical architecture

Deployment is split across DC-EU-DUB (primary) and DC-US-RDU-COLO (DR):

Role	Hosts	Spec
Argus App	4 x Windows Server 2012 R2	16 vCPU, 64 GB RAM each
Oracle 11g RAC	2-node cluster	48 vCPU, 384 GB RAM, Fibre Channel SAN 5 TB
JBoss gateway	2 x RHEL 6	8 vCPU, 32 GB RAM each
Siebel portal	2 x Oracle Linux 6	16 vCPU, 48 GB RAM each
Dashboard web	2 x Windows Server 2016	8 vCPU, 16 GB RAM each
Literature workers	4 x RHEL 7	8 vCPU, 16 GB RAM each

DR failover was last successfully tested in May 2025. The November 2025 drill failed when the DR Argus app servers were missing a 2024 security patch that broke E2B generation on failover.

6. Technology stack

Component	Version	Vendor support status
Oracle Argus Safety	7.0.2 (customised)	**End of standard support** (Oracle recommends 8.4+); extended support contract active until 2027
Oracle Database Enterprise	11g R2	**End of life** (extended support ended Dec 2020; sustaining-support only, no patches)
Oracle Siebel CRM	8.1	**End of life** (support ended Dec 2019)
Microsoft Windows Server	2012 R2	**End of life** (ended Oct 2023)
JBoss EAP	6.4	**End of life** (ended Jun 2022)
Red Hat Enterprise Linux	6	**End of life** (maintenance support ended Jun 2024)
Oracle Linux	6	**End of life**
ASP.NET MVC	5 (.NET 4.6.2)	Supported but frozen; .NET Framework 4.6.2 reached EOL Apr 2022
Python	3.6 (literature workers)	**End of life** (Dec 2021)
Internet Explorer 11	n/a	**Retired** (Microsoft support ended Jun 2022); staff workaround is Edge IE Mode

SCA report (October 2026): 62 critical and 189 high CVEs across the full dependency inventory. The JBoss and Oracle 11g findings are the highest priority.

7. Integrations

- **FDA ESG, EMA EV, PMDA, Health Canada, TGA, Swissmedic, ANVISA, CDSCO +12 other agencies** via E2B(R3) over AS2 and SOAP endpoints
- **Partner licensees (6 pharma companies)** — bi-directional ICSR exchange via E2B(R3) SFTP drops
- **Vernova Clinical Safety (ClinSafe)** — nightly bi-directional reconciliation ETL
- **Vernova Medical Information** — inbound case feed (customer-reported adverse events from the call centre)
- **Embase + PubMed APIs** — literature screening
- **MedDRA MSSO** — semi-annual MedDRA term update (manual process)
- **Corporate SSO (Active Directory + MFA via RSA SecurID)** — identity

The AS2 submission path to FDA ESG is the most sensitive integration. A failed submission that is not detected within 24 hours is a reportable regulatory event in itself.

8. Non-functional characteristics

Performance: Argus thick-client is adequate at the EU and US sites. India and Japan sites report >8 second case-open latency during month-end PBRER cycles.

Availability: 99.78% over the past 12 months (SLA target 99.9% — missed). Two outages during Day-15 submission windows triggered formal regulatory incident reports.

Scalability: Oracle 11g RAC is the bottleneck. ICSR volume is growing 12% per year and will exceed current throughput by FY29 without intervention.

DR: Oracle Data Guard SYNC replication is working. The DR app-tier gap identified in the November 2025 drill has been remediated but DR has not been re-tested since.

9. Operations and support

- 24x7 managed service by an external vendor (contract renewed 2024, expires 2027).
- Deployment: 2-3 months. 180+ page validation deliverable per release.
- Testing: approximately 650 OQ test cases, approximately 60% automated via a vendor test automation suite.
- Monitoring: enterprise Splunk + bespoke E2B submission dashboard.
- Backups: Oracle RMAN nightly to tape + daily snapshot to secondary SAN. 25-year retention.

10. Compliance posture

Validation status

Validated under GAMP 5 Category 4 (Argus itself) with Category 5 modules (custom gateway, custom dashboards, literature workflow). URS, FS, DS last amended June 2026.

21 CFR Part 11

- Audit trail: implemented in Argus natively on 78 tables, extended to custom modules.
- E-signatures: on case-lock, case-unlock, assessment overrides. Backed by RSA SecurID MFA.
- Access controls: role-based, reviewed semi-annually by QA.
- MFA: enforced at login AND at e-signature events (unlike LabTrack).

GVP Module I compliance posture

- EU-QPPV named: Dr. Alain Dubois.
- Deputy QPPV: Dr. Fatima El-Amrani.
- PSMF (Pharmacovigilance System Master File) maintained in a separate document system; SafetyNet is referenced throughout.

Open audit findings

ID	Severity	Open since	Description
PV-OBS-2024-012	High	Mar 2024	Submission gateway lacks automated detection of silent AS2 failures (reportable incident risk)
GXP-OBS-2025-101	High	Oct 2025	Oracle 11g R2 is sustaining-support only; no security patches available
EMA-INS-2026-003	Critical	Mar 2026	EMA Pharmacovigilance Inspection: lack of real-time monitoring for ACK3 negative acknowledgements
PV-OBS-2026-027	Medium	Aug 2026	DR Argus app tier not re-tested after November 2025 drill failure

EMA-INS-2026-003 is a formal inspection finding with a committed remediation date of 31 March 2027. This is the hard deadline the modernization plan must respect.

Signal detection and AI

Signal detection uses Argus's native tools (PRR, ROR, information component). The 2023 AI-assisted narrative drafter is implemented using Vernova's enterprise LLM gateway — review process is approved by QA; however, QA has not yet approved its use for GVP-reportable cases, only for internal medical review drafts.

11. Pain points

- **EU-QPPV (Dr. Dubois):** "If a submission fails silently I am personally liable. The EMA finding is non-negotiable. Any plan must deliver real-time gateway monitoring by Q1 FY27."
- **Head of Patient Safety:** "ICSR volume will exceed what the current platform can handle by FY29. We cannot allow a scalability-driven incident."
- **IT Service Owner (vendor):** "We need direction on Argus — stay on Argus 8.4 (incremental upgrade), move to another vendor (Veeva Vault Safety, Oracle Argus Cloud), or build. The managed-service contract renewal is in 2027."
- **Information Security:** "Oracle 11g R2 has had unpatched kernel-level CVEs for 18 months. The compensating controls are stretched."
- **Deputy QPPV:** "The literature screening workflow is 2010s technology. A Python 3.6 Celery farm is embarrassing."

12. Prior modernization attempts

- **2022:** POC on Veeva Vault Safety reached a working ICSR flow but was halted when the business case assumed too-aggressive user adoption and FY23 budget cut the project.

- **2024:** Argus 8.4 upgrade POC completed in sandbox; blocked by concerns about customisation portability (the custom gateway code depends on Argus 7.0 internal APIs that changed in 8.x).

13. Constraints

- **Non-negotiable deadline: 31 March 2027** for the EMA-INS-2026-003 finding — gateway monitoring in production.
- **Validation state must be continuous** — pharmacovigilance cannot run in an unvalidated state even briefly. Shadow-mode and parallel-run approaches only.
- **ICSR data residency:** EU subject data in the EU; US in the US; India in India (per DPDP + CDSCO guidance under consultation).
- **MedDRA dictionary refresh cadence** must be maintained semi-annually without disrupting live operations.
- **No downtime during Day-15 submission windows** (the 5-day pre-deadline window for each product).
- **Audit trail continuity for 25 years** per EMA retention guidance.
- **Budget envelope:** €4.5M to €5.5M across FY27 and FY28.
- **The managed-service contract expires December 2027** — must be renewed, replaced, or in-sourced by then.
- **24x7 on-call must be preserved** throughout migration.

14. Decision points for the modernization engagement

- **Target platform**: incremental Argus 8.x upgrade; migrate to Oracle Argus Cloud; migrate to Veeva Vault Safety; or (unlikely) in-house build.
- **Vendor managed-service**: renew, replace, or in-source.
- **AI drafter scope**: expand to GVP-reportable cases (requires QA sign-off and validation of the AI assurance process).
- **Gateway rebuild**: replace JBoss custom gateway with a cloud-managed AS2/SOAP gateway, OR extend the existing Argus-included gateway.
- **Partner licensee data exchange**: continue SFTP + E2B, or move to a managed B2B integration platform.

15. Expected outcomes of modernization

- Meet the 31 March 2027 EMA deadline for real-time gateway monitoring.
- Exit Oracle 11g R2 and all other EOL technology.
- Deliver scalability headroom to 100,000 ICSRs per year.
- Reduce release cycle from 2-3 months to monthly.
- Deliver a cost reduction of 20 to 30% versus current €2.1M annual run cost.
- Enable AI-assisted narrative drafter for GVP-reportable cases (pending QA process approval).

- Preserve 24x7 operations and validated state throughout.